

LiveScribe

Your Conversation, Understood in Real Time — On-Device Snapdragon® AI

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GitHub: <https://github.com/SlayerHADES/LiveScribe>

- **100% Private:** Audio & transcripts never leave the device.
- **Sub-300ms Latency:** Real-time streaming Whisper speech recognition.
- **Zero Internet Required:** Fully functional in Airplane Mode on the Hexagon NPU.

1. The Problem

- **Cloud Dependency:** Existing tools (Otter, Fathom, Fireflies) upload raw microphone audio to remote GPU servers.
- **Privacy Risks:** Executive meetings, financial reviews, and IP research exposed to cloud leaks.
- **Connectivity Failure:** Cloud speech tools fail completely in flight, in basement labs, or in low-signal areas.

2. The Solution — LiveScribe

- **On-Device Hexagon NPU Processing:** End-to-end execution of STT and LLM summarization locally via ONNX Runtime QNN Provider.
- **Real-Time Visualizer & Streaming:** Canvas audio visualizer with word-by-word streaming text animation.
- **Automated Meeting Intelligence:** Instant generation of executive briefs, decision lists, and task owner tables.

3. Technical Architecture

Pipeline Flow:

Microphone Audio (16kHz PCM) → WebSocket Stream → ONNX Runtime QNN Provider → Hexagon NPU → Whisper Transcript → Local LLM Summarization

- **Backend:** Python FastAPI, WebSocket streaming (`WS /ws/transcribe`)
- **Frontend:** Venture-backed SaaS studio UI (Linear + Raycast + Chatbase + Reevo aesthetic) with HTML5 Canvas visualizer and Cmd+K command palette.

4. Why NPU — Performance Benchmarks

Metric	Cloud Tools	LiveScribe (Hexagon NPU)
Connectivity	Requires Active Internet	100% Offline (Airplane Mode)
Data Privacy	Sent to Remote Cloud	0 Bytes Leave Laptop
Hardware	Remote GPU Cluster	Qualcomm Hexagon NPU
Chunk Latency	1000 - 3000 ms	~300 ms / chunk

5. Interactive AI & Notes Export

- **Timestamp Citing AI:** Ask AI assistant cites clickable timestamps (■ 10:43 — Kshitij M.) auto-scrolling to exact transcript lines.
- **Raycast Command Palette:** Cmd+K search modal with instant query match highlighting.
- **One-Click Markdown Export:** Generates complete meeting notes document (LiveScribe_Meeting_Notes.md).